

Notes: Rice and Strahan (JF, 2010)
Does Credit Competition Affect Small-Firm Finance?

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1 Big picture

- **Research question.** Does stronger credit competition affect how small firms borrow? More specifically, when states are more open to interstate bank branching, do small firms face lower loan rates, borrow more from banks, receive better nonprice loan terms, or change their capital structure?
- **Main idea.** The paper uses state-level restrictions on interstate branching after the Interstate Banking and Branching Efficiency Act (IBBEA) as variation in bank competition. Some states made out-of-state entry easy, while others kept barriers that limited de novo branching, single-branch acquisitions, and deposit concentration.
- **Empirical object.** The core explanatory variable is a state-year *branching restrictions index* ranging from zero to four. Zero means the state is most open to out-of-state branching; four means it is most restrictive.
- **Main finding on prices.** Small firms in states more open to interstate branching pay lower rates on loans. Comparing the most open states to the most restrictive states implies roughly 80 to 100 basis points lower interest rates.
- **Main finding on lender choice.** Firms in more open states are more likely to borrow from banks, consistent with competition expanding the supply of bank credit.
- **Main finding on quantities.** Lower loan rates do not translate into more total borrowing, higher approval rates, or clearly looser nonprice loan terms.
- **Interpretation.** Credit competition lowers prices, but for small opaque firms it does not necessarily relax quantity constraints. Banks may still ration credit or rely on collateral, guarantees, and maturity restrictions because contracting problems remain.
- **Key surprise.** The paper finds a large price effect but little quantity effect. This distinguishes competition-driven supply changes from liquidity-driven supply shocks, which often show stronger effects on the amount borrowed.

Economic takeaway. More bank competition makes bank loans cheaper and shifts small firms toward banks, but it does not mechanically increase leverage or eliminate credit constraints. Competition changes the price of credit more clearly than the quantity of credit.

2 Incremental contribution of the paper

- **A clean competition shock.** The paper studies a regulatory change that alters the competitive structure of banking markets. This differs from credit-supply papers that use bank capital, liquidity, or monetary-policy shocks.
- **Uses cross-state implementation of IBBEA.** IBBEA opened interstate branching nationally, but states retained discretion over several entry barriers. This creates useful cross-state and time-series variation after 1997.
- **Focuses on small private firms.** The Survey of Small Business Finance (SSBF) covers firms with fewer than 500 employees. These firms are especially dependent on banks and are unlikely to have access to public bond or equity markets.

- **Directly observes loan prices and terms.** The SSBF reports the interest rate on the most recent loan, collateral, guarantees, maturity, denied/discouraged borrowing, bank debt use, and trade credit. This allows the paper to distinguish price effects from quantity and contract-term effects.
- **Falsification using 1993 data.** The authors assign the later branching index to the pre-IBBEA 1993 survey. The branching index does not predict loan rates in 1993, supporting the view that post-IBBEA results are not simply cross-state demand differences.
- **Connects bank competition to capital structure.** The paper asks whether lower loan prices are enough to change leverage and debt composition. The answer is mostly no, which is informative about small-firm credit constraints.
- **Political-economy controls.** The paper studies whether small-bank power, insurance-sector size, political control, and local growth explain state restrictions. The loan-rate result remains after controlling for these potential determinants.
- **Clarifies the mechanism.** The evidence suggests that interstate branching expands credit supply by increasing lender competition, but that asymmetric-information and relationship-lending frictions may still bind.

3 Data and measurement

3.1 Institutional setting: IBBEA and state branching barriers

The Interstate Banking and Branching Efficiency Act of 1994 allowed bank holding companies to expand across state lines and allowed interstate branching beginning in 1997. However, states retained authority to restrict the way out-of-state banks could enter.

- **Age restrictions.** States could require a target bank to have existed for a minimum number of years before it could be acquired in an interstate merger. The paper adds one to the index if the minimum age is at least three years.
- **De novo interstate branching.** States had to explicitly permit out-of-state banks to open new branches. The paper adds one if the state does not allow de novo interstate branching.
- **Single-branch acquisition.** States had to explicitly allow an out-of-state bank to acquire individual branches rather than an entire bank. The paper adds one if the state does not allow this.
- **Statewide deposit cap.** IBBEA imposed a default 30% deposit cap but allowed states to use tighter limits. The paper adds one if the cap is below 30%.

The resulting index is:

$$\text{Branching Restrictions}_{jt} \in \{0, 1, 2, 3, 4\}, \quad (1)$$

where a higher value means more barriers to entry and weaker potential interstate competition.

Interpretation. The index is not a direct measure of realized competition; it measures the regulatory barriers that shape entry by large out-of-state banks. If entry barriers bind, higher values should be associated with weaker competition and higher borrowing costs.

3.2 Small-firm data

The firm-level data come from the Survey of Small Business Finance. The relevant survey years are 1993, 1998, and 2003.

- **Firm type.** The survey covers firms with fewer than 500 employees.
- **Repeated cross sections.** The survey is not a panel. Each survey year contains a different sample of firms.
- **Pre-treatment year.** The 1993 survey is before the effective interstate branching treatment and is used as a falsification/control sample.
- **Post-treatment years.** The 1998 and 2003 surveys are after IBBEA's branching provisions became effective.
- **Survey weights.** The regressions use SSBF weights to account for disproportionate sampling and unit nonresponse.

Interpretation. The repeated cross-section structure means the paper cannot track individual firms before and after reform. Instead, it compares similar firms operating in states with different regulatory openness.

3.3 Key outcomes

The paper studies three groups of outcomes.

- **Loan prices.** The main dependent variable is the interest rate on the most recent loan. This variable is available for about 40% of the sample.
- **Credit quantities and access.** Outcomes include whether the firm has bank debt, log total debt, whether it was denied or discouraged from applying for a loan, and late trade-credit payments.
- **Nonprice terms.** Outcomes include whether the loan has collateral, whether the loan is guaranteed, and loan maturity.

Interpretation. Observing prices, quantities, and terms is crucial. A pure credit-demand shock should tend to raise both prices and quantities. A supply/competition shock may lower prices, but quantity effects depend on contracting frictions and rationing.

3.4 Control variables

The interest-rate regressions follow the small-business lending literature and include a rich set of controls.

- **Interest-rate controls.** Prime rate, term spread, and default spread at the time of loan approval.
- **Lender controls.** Indicators for bank lender and nonfinancial-firm lender.
- **Loan controls.** Indicator for floating-rate loans.
- **Borrower controls.** Log assets, age, ROA, corporation status, credit risk rating, and two-digit SIC industry indicators.
- **Relationship controls.** Length of the lender-borrower relationship, number of lender relationships, deposit relationship, and information/noninformation services from the lender.
- **Market controls.** Local deposit-market concentration, urban/MSA indicator, and lagged 5-year local income growth.

Interpretation. These controls help compare firms with similar risk, size, relationship histories, loan characteristics, and local credit-market environments.

4 Methodology and empirical specifications

4.1 Year-by-year interest-rate regressions

The benchmark loan-rate specification is estimated separately for 1993, 1998, and 2003:

$$Y_{ijt} = \beta_t \text{Branching Restrictions}_{jt} + \text{Interest-rate, lender, firm, and market controls}_{ijt} + \varepsilon_{ijt}, \quad (2)$$

where i indexes firms, j indexes states, and t indexes survey years. The dependent variable Y_{ijt} is the interest rate on the firm's most recent loan.

- The coefficient β_t measures how loan rates vary with the restrictiveness of interstate branching rules in that year.
- The 1993 coefficient is a falsification check, since the interstate branching treatment had not yet occurred.
- Standard errors are clustered by state in the year-by-year regressions.

Interpretation. A positive coefficient means firms in more restrictive states pay higher rates. Since the index ranges from zero to four, multiplying the coefficient by four gives the approximate difference between most-open and most-restricted states.

4.2 Pooled model with state fixed effects

The pooled model uses all three SSBF years and includes state and year fixed effects:

$$Y_{ijt} = \alpha_t + \gamma_j + \beta \text{Branching Restrictions}_{jt} + \text{Interest-rate, lender, firm, and market controls}_{ijt} + \varepsilon_{ijt}. \quad (3)$$

In this specification, the branching index is coded as four for all states in 1993, because before interstate branching every state is treated as fully restricted.

- **Year effects** absorb common macroeconomic and interest-rate conditions.
- **State effects** absorb permanent differences across states.
- Identification comes from changes over time in states' openness to interstate branching.

Interpretation. The state fixed effects specification is stronger than the cross-sectional post-IBBEA comparison because it removes time-invariant state characteristics.

4.3 Political-economy and growth controls

The paper acknowledges that state rules were not randomly chosen. States with strong small-bank interests or certain growth conditions may have been more likely to restrict out-of-state branching.

The authors therefore study the correlation between the branching index and several state-level variables:

- small-bank share of banking assets;
- insurance-sector size;
- small-firm share of firms in the state;
- capital ratio of small banks relative to large banks;
- governor and legislature political-party variables;
- lagged state income growth.

They then add these variables one at a time to the interest-rate regressions.

Interpretation. This is not a perfect randomization test, but it checks whether the main price effect is simply driven by observable determinants of state regulatory choices.

4.4 Credit quantities and nonprice terms

After showing that interstate branching openness lowers loan rates, the paper asks whether this translates into broader credit access.

The pooled models use outcomes such as:

Bank debt indicator, $\log(\text{Total debt})$, Denied/discouraged indicator, Late trade credit, (4)

and nonprice terms:

Collateral indicator, Guarantee indicator, Loan maturity. (5)

- Probit models are used for some binary outcomes, with marginal effects reported.

- The regressions include the same broad controls as the loan-rate models.
- Standard errors are clustered by state-year.

Interpretation. These tests ask whether competition only reduces rates or also changes the amount and contractual structure of credit.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies how small-firm credit outcomes vary with state openness to interstate branching, after controlling for firm, loan, lender, market, industry, state, and year characteristics. The strongest evidence is for loan prices.

The paper does not fully randomize competition. Instead, it relies on the institutional design of IBBEA, cross-state differences in state choices, pre-treatment falsification evidence, and robustness checks using political-economy controls.

5.2 Why the 1993 falsification matters

If the branching index were merely proxying for permanent state-level credit-demand conditions, it should predict loan rates even before interstate branching took effect. The 1993 coefficient is essentially zero, while the post-IBBEA coefficients are positive and significant.

Economic meaning. This pattern supports the view that the branching restrictions affected credit supply after interstate entry became relevant, rather than simply capturing long-standing differences across states.

5.3 Price versus quantity

The paper's main empirical tension is that prices respond strongly but quantities respond weakly.

- **Price response.** More open states have lower loan rates.
- **Bank-use response.** Firms are more likely to have bank debt when restrictions are lower.
- **Quantity nonresponse.** Total debt, denial/discouragement, and maturity do not show strong improvements.

Interpretation. Competition may make banks price more aggressively, but it need not solve adverse selection, moral hazard, or information problems that limit small-firm borrowing.

5.4 Why the nonresults are informative

The absence of a large quantity response helps the interpretation in two ways.

- It weakens the concern that the branching index is proxying for higher credit demand. Higher demand should tend to raise borrowing quantities.
- It suggests that different credit-supply shocks operate through different margins. Liquidity shocks may affect quantities strongly, while competition shocks may affect prices more strongly.

Economic meaning. The results imply that “credit supply” is not one-dimensional. A supply expansion due to greater competition can lower rates without necessarily increasing leverage.

5.5 Relationship lending and possible rationing

Competition can reduce bank market power, which lowers prices. But competition can also weaken relationship rents and information production. If banks have less ability to earn future rents from a borrower, they may be less willing to absorb short-run information or monitoring costs.

The paper therefore leaves room for the possibility that competition lowers rates for loans that are made but does not relax, and may even worsen, some forms of credit rationing.

6 Tables and figures

6.1 Table I and Figure 1: State branching laws and the timing of reform

Read Table I:

- Table I lists each state’s interstate branching restrictions from 1994 to 2005.
- The index is built from four barriers: minimum age requirements, no de novo branching, no single-branch acquisition, and deposit caps below 30%.
- States such as Maryland, Michigan, North Carolina, Ohio, Pennsylvania, Rhode Island, and Virginia are very open in the relevant period.
- States such as Arkansas, Colorado, Iowa, Kansas, Mississippi, Missouri, and Nebraska are among the most restrictive.

Economic message: The table shows that IBBEA did not create a uniform national banking market. State-level policy choices preserved meaningful variation in entry barriers.

Table I
State Interstate Branching Laws: 1994–2005

This table lists the index of interstate branching restrictions, the effective date of interstate branching regulation changes, and each of the following four provisions: the minimum age of the institution for acquisition, allowance of de novo interstate branching, allowance of interstate branching by acquisition of a single branch or portions of an institution, and statewide deposit cap on branch acquisitions. The index is set to zero for states that are most open to out-of-state entry. We add one to the index when a state adds any of the four barriers just described. Specifically, we add one to the index: if a state imposes a minimum age of 3 or more years on target institutions of interstate acquirers; if a state does not permit de novo interstate branching; if a state does not permit the acquisition of individual branches by an out-of-state bank; and if a state imposes a deposit cap less than 30%. The index ranges from zero to four. Source: Johnson and Rice (2008).

State	Branching Restrictiveness Index	Effective Date	Minimum Age of Institution (Bank or Branch) for Acquisitions	Allows de novo Interstate Branching	Interstate Branching by Acquisition of Single Branch or Portions of an Institution	Statewide Deposit Cap on Branch Acquisitions
Alabama	3	5/31/1997	5 years	No	No	30%
Alaska	2	1/1/1994	3 years	No	Yes	50%
Arizona	2	8/31/2001	5 years	No	Yes	30%
Arizona	3	9/1/1996	5 years	No	No	30%
Arkansas	4	6/1/1997	5 years	No	No	25%
California	3	9/28/1996	5 years	No	No	30%
Colorado	4	6/1/1997	5 years	No	No	25%
Connecticut	1	6/27/1995	5 years	Yes	Yes	30%
Delaware	3	9/29/1995	5 years	No	Yes	30%
DC	0	6/13/1996	No	Yes	Yes	30%
Florida	3	6/1/1997	3 years	No	No	30%
Georgia	3	5/10/2002	3 years	No	No	30%
Georgia	3	6/1/1997	5 years	No	No	30%
Hawaii	0	1/1/2001	No	Yes	Yes	30%
Hawaii	3	6/1/1997	5 years	No	No	30%
Idaho	3	9/29/1995	5 years	No	No	None
Illinois	0	8/20/2004	No	Yes	Yes	30%
Illinois	3	6/1/1997	5 years	No	No	30%
Indiana	1	7/1/1998	5 years	Yes	Yes	30%
Indiana	0	6/1/1997	No	Yes	Yes	30%
Iowa	4	4/4/1996	5 years	No	No	15%
Kansas	4	9/29/1995	5 years	No	No	15%
Kentucky	3	3/22/2004	No	No	No	15%
Kentucky	3	3/17/2000	No	No	No	15%
Kentucky	4	6/1/1997	5 years	No	No	15%
Louisiana	3	6/1/1997	5 years	No	No	30%
Maine	0	1/1/1997	No	Yes	Yes	30%
Maryland	0	9/29/1995	No	Yes	Yes	30%
Massachusetts	1	8/2/1996	3 years	Yes	Yes	30%
Michigan	0	11/29/1995	No	Yes	Yes	None
Minnesota	3	6/1/1997	5 years	No	No	30%
Mississippi	4	6/1/1997	5 years	No	No	25%
Missouri	4	9/29/1995	5 years	No	No	13%
Montana	4	10/1/2001	5 years	No	No	22%

Table I, part 1: State interstate branching laws.

Table I—Continued

State	Branching Restrictiveness Index	Effective Date	Minimum Age of Institution (Bank or Branch) for Acquisitions	Allows de novo Interstate Branching	Interstate Branching by Acquisition of Single Branch or Portions of an Institution	Statewide Deposit Cap on Branch Acquisitions
Montana	4	9/29/1995	N/A	N/A	N/A	Increases 1% per year from 18% to 22%
Nebraska	4	5/31/1997	5 years	No	No	14%
Nevada	3	9/29/1995	5 years	Limited	Limited	30%
New Hampshire	0	1/1/2002	No	Yes	Yes	30%
New Hampshire	1	8/1/2000	5 years	Yes	Yes	30%
New Hampshire	4	6/1/1997	5 years	No	No	20%
New Jersey	1	4/17/1996	No	No	Yes	30%
New Mexico	3	6/1/1996	5 years	No	No	40%
New York	2	6/1/1997	5 years	No	Yes	30%
North Carolina	0	7/1/1995	No	Yes	Yes	30%
North Dakota	1	8/1/2003	No	Yes	Yes	25%
North Dakota	3	5/31/1997	No	No	No	25%
Ohio	0	5/21/1997	No	Yes	Yes	30%
Oklahoma	1	5/17/2000	No	Yes	Yes	20%
Oklahoma	4	5/31/1997	5 years	No	No	15%
Oregon	3	7/1/1997	3 years	No	No	30%
Pennsylvania	0	7/6/1995	No	Yes	Yes	30%
Rhode Island	0	6/20/1995	No	Yes	Yes	30%
South Carolina	3	7/1/1996	5 years	No	No	30%
South Dakota	3	3/9/1996	5 years	No	No	30%
Tennessee	1	3/17/2003	3 years	Yes	Yes	30%
Tennessee	1	7/1/2001	5 years	Yes	Yes	30%
Tennessee	2	5/1/1998	5 years	No	Yes	30%
Tennessee	3	6/1/1997	5 years	No	No	30%
Texas	2	9/1/1999	No	Yes	Yes	20%
Texas	4	8/28/1995	N/A	N/A	N/A	20%
Utah	1	4/30/2001	5 years	Yes	Yes	30%
Utah	2	6/1/1995	5 years	No	Yes	30%
Vermont	0	1/1/2001	No	Yes	Yes	30%
Vermont	2	5/30/1996	5 years	No	Yes	30%
Virginia	0	9/29/1995	No	Yes	Yes	30%
Washington	1	5/9/2005	5 years	Yes	Yes	30%
Washington	3	6/6/1996	5 years	No	No	30%
West Virginia	1	5/31/1997	No	Yes	Yes	25%
Wisconsin	3	5/1/1996	5 years	No	No	30%
Wyoming	3	5/31/1997	3 years	No	No	30%

Table I, part 2: Continued state interstate branching laws.

Read Figure 1:

- The timeline places interstate branching in the broader history of bank deregulation.
- Intrastate branching deregulation began earlier, followed by interstate banking through bank holding companies.
- IBBEA was passed in 1994, and interstate branching became effective in 1997.
- The 1993 SSBF survey therefore provides a pre-treatment control, while the 1998 and 2003 surveys provide post-treatment observations.

Economic message: The figure clarifies the timing of the empirical design. The key treatment is interstate branching after 1997, not the earlier intrastate deregulation wave.

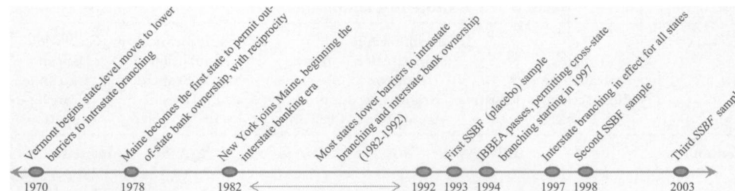


Figure 1. Timeline of intra- and interstate branching deregulation, 1970–2003.

Figure 1: Timeline of intra- and interstate branching deregulation, 1970–2003.

6.2 Table II: Summary statistics

Read:

- Table II reports summary statistics for loan terms, interest rates, borrower characteristics, lender characteristics, relationship variables, and local market characteristics.
- The average loan rate is 8.47% in 1993, 9.04% in 1998, and 5.79% in 2003.
- The average branching restrictions index is 1.99 in 1993, 2.43 in 1998, and 2.05 in 2003.
- Borrowers are small and relationship-intensive. The average lender-borrower relationship is 8.07 years in 1993, 5.48 years in 1998, and 10.37 years in 2003.
- Bank lenders dominate the most recent loan sample, especially in 1993 and 2003.

Economic message: The table shows that small-firm credit markets are relationship-based and bank-centered, making them a natural setting for studying bank competition.

Table II Summary Statistics						
This table reports summary statistics from the 1993, 1998, and 2003 Surveys of Small Business Finance. Each survey contains a random sample of firms with fewer than 500 employees. Each sample is drawn independently, so we do not follow the same firms over time. The data on loan terms, interest rates, loan characteristics, and lender characteristics reflect information from the most recent loan by the borrower; these data are available for about 40% of the sample. The other statistics are available for all of the respondents, but here we report the summary statistics for the smaller sample that is included in the interest rate regressions (Tables III and V). Market concentration equals the sum of squared share of deposits held by all banks in the borrower's local market, where market is defined as the Metropolitan Statistical Area (MSA) or county.						
	1993		1998		2003	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
<i>Loan terms</i>						
Interest rate on most recent loan	8.47	2.21	9.04	2.37	5.79	2.68
Share with collateral	0.72	0.45	0.63	0.48	0.55	0.50
Share guaranteed	0.56	0.50	0.56	0.50	0.60	0.49
Loan maturity (years)	3.28	4.38	4.57	5.49	3.76	4.95
<i>Index of branching restrictions</i>						
4 is most, 0 least, restricted	1.99	1.35	2.43	1.41	2.05	1.34
<i>Interest rates</i>						
Prime interest rate	6.00	0.00	8.35	0.27	4.13	0.12
Term structure (10-year government bond – 3-month T-bill)	2.84	0.36	0.35	0.14	2.94	0.39
Default premium (BAA – 10-year government bond)	4.91	0.42	2.31	0.35	2.75	0.29
<i>Borrower characteristics</i>						
Log of borrower assets	13.25	2.11	12.75	2.25	13.51	2.13
Indicator if borrower is a corporation	0.46	0.50	0.33	0.47	0.33	0.47
Borrower ROA	0.33	0.65	0.66	1.10	0.46	0.82
Borrower risk rating (1 is safest; 5 is riskiest)	2.94	1.16	3.00	1.14	2.74	1.18
<i>Loan characteristics</i>						
Indicator if loan is floating-rate	0.58	0.49	0.34	0.47	0.55	0.50
<i>Lender characteristics</i>						
Indicator if lender is a bank	0.88	0.33	0.77	0.42	0.87	0.34
Indicator if lender is a nonfinancial company	0.03	0.16	0.05	0.21	0.02	0.14
Indicator if loan is a line of credit renewal		N/A			0.46	0.50
<i>Relationship variables</i>						
Length of lender–borrower relationship (years)	8.07	8.34	5.48	7.09	10.37	10.56
Borrower age (years)	16.27	14.24	13.90	10.95	18.32	13.08
Number of information services	0.26	0.44	0.17	0.37	0.39	0.49

Table II, part 1: Summary statistics.

Table II—Continued						
	1993		1998		2003	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
Number of noninformation services from lender	0.26	0.44	0.32	0.47	0.48	0.50
Indicator if borrower has a deposit with lender	0.72	0.45	0.53	0.50	0.75	0.43
Number of relationships	1.94	1.44	2.10	1.71	2.19	1.57
<i>Market characteristics</i>						
Local market concentration	0.16	0.09	0.17	0.09	0.17	0.10
Indicator if borrower is in an MSA	0.77	0.42	0.74	0.44	0.93	0.26
Lagged 5-year growth rate in the local market	0.22	0.06	0.25	0.06	0.19	0.06

Table II, part 2: Continued summary statistics.

6.3 Table III: Branching restrictions and loan interest rates

Read:

- The dependent variable is the interest rate on the most recent loan.

- In 1993, before the treatment, the branching restrictions coefficient is close to zero and insignificant.
- In 1998, the coefficient is 0.21 with a t -statistic of 3.01.
- In 2003, the coefficient is 0.25 with a t -statistic of 3.62.
- In the pooled state-fixed-effects model, the coefficient is 0.22 with a t -statistic of 3.84.
- Since the index ranges from zero to four, these estimates imply about 80 to 100 basis points lower loan rates in the most open states relative to the most restrictive states.

Economic message: This is the central price result. Interstate branching openness lowers the cost of small-business borrowing.

Table III
Regression of Interest Rate on Most Recent Loan on State, Bank, and Borrower Characteristics

This table reports regressions of the interest rate on the most recent loan on state, bank, and borrower characteristics. The regressions are weighted by the survey weights, which account for disproportionate sampling and unit nonresponse, and include a set of two-digit SIC indicator variables to control for industry effects. Standard errors are clustered by state (state-year in the pooled model). The pooled model includes both state and year fixed effects. In the 1993 sample, we assign the 2003 value for the branching index; in the pooled model, in contrast, we assign the value of four to all states in 1993 since this is the most restrictive value that the variable takes. Market concentration equals the sum of squared share of deposits held by all banks in the borrower's local market, where market is defined as the Metropolitan Statistical Area (MSA) or county.

	1993 (Control Group)		1998		2003		Pooled, with State Fixed Effects	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
Index of branching restrictions (4 is most, 0 least, restricted)	0.01	0.15	0.21	3.01	0.25	3.62	0.22	3.84
<i>Interest rates</i>								
Prime interest rate	N/A		-2.70	3.05	-1.12	0.35	-0.84	2.23
Term structure (10-year government bond - 3-month Tbill)	-0.39	0.28	0.25	0.27	0.43	0.72	0.02	0.11
Default premium (BAA - 10-year government bond)	0.11	0.10	-1.68	2.95	0.63	0.77	-0.09	0.66
<i>Borrower characteristics</i>								
Lag of borrower assets	-0.23	4.12	-0.25	3.26	-0.27	5.62	-0.28	7.78
Indicator if borrower is a corporation	-0.18	1.39	-0.30	1.08	-0.34	1.69	-0.17	1.60
Borrower ROA	-0.23	2.34	0.02	0.18	0.01	0.04	-0.06	0.71
Borrower risk rating (1 is safest; 5 is riskiest)	0.14	2.17	0.08	1.00	0.24	2.59	0.19	3.19
<i>Loan characteristics</i>								
Indicator if loan is floating-rate	-0.51	3.23	-0.17	0.74	-0.94	4.55	-0.61	5.00
<i>Lender characteristics</i>								
Indicator if lender is a bank	-0.26	0.47	-0.66	1.29	-1.13	2.33	-0.75	2.48
Indicator if lender is a nonfinancial company	-0.30	0.38	-0.21	0.27	-1.13	0.67	-0.75	1.01

(continued)

Table III, part 1: Loan-rate regressions.

6.4 Tables IV and V: Are branching restrictions proxying for state characteristics?

Read Table IV:

- Table IV reports correlations between the branching index and possible determinants of state policy.
- Small-bank share is positively correlated with the branching index, consistent with small banks favoring restrictions on out-of-state entry.
- Lagged state income growth from 1993 to 1997 is also positively correlated with the index.
- Political-party variables and insurance-sector size are not strongly related to the index.

Economic message: Branching restrictions are partly political-economy outcomes, especially related to small-bank interests. This motivates robustness checks.

Table III—Continued

	1993 (Control Group)		1998		2003		Pooled, with State Fixed Effects	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
<i>Relationship variables</i>								
Length of lender-borrower relationship (years)	0.00	0.33	-0.03	2.54	0.01	1.13	0.00	0.66
Borrower age (years)	0.00	0.29	-0.01	1.19	-0.02	2.51	-0.01	2.05
Number of information services from lender	-0.47	2.71	-0.34	0.99	0.10	0.45	-0.13	0.85
Number of noninformation services from lender	0.07	0.39	-0.14	0.45	0.10	0.53	0.00	0.94
Indicator if borrower has a deposit with lender	-0.37	2.12	0.18	0.74	0.30	1.27	0.11	0.78
Number of relationships	0.09	1.36	0.07	1.02	0.06	0.89	0.04	0.94
<i>Market characteristics</i>								
Market concentration	0.39	0.40	-1.59	1.11	0.78	0.83	0.42	0.58
Indicator if borrower is in an MSA	0.06	0.32	-0.54	1.91	-0.26	0.70	-0.08	0.69
Average lagged 5-year growth rate	-0.45	0.33	0.98	0.45	-0.46	0.18	0.46	0.50
N	1,636		788		1,737		4,161	
R^2	16.31%		20.06%		23.51%		31.65%	

Table III, part 2: Continued loan-rate regressions.

Table IV
Correlation between Interest Group Factors, Political Factors, and Economic Conditions with Index of Branching Restrictions

For each variable described in the far left column, this table reports its mean and its correlation with the branching restrictions index, where each statistic is built from 51 observations, one per state (including DC). With the exception of the economic growth variables, all state characteristics are measured as of 1993, the year before passage of the Interstate Banking and Branching Efficiency Act. **Denotes significance at the 1% level.

	Mean	Cross-State Correlation with Branching Index	
		1998 Index	2003 Index
Small bank share of all banking assets in state	0.059	0.49**	0.39**
Relative size of insurance to banking plus insurance	0.443	-0.16	-0.22
Small firm share of the number of firms in the state	0.761	0.16	0.15
Capital ratio of small banks relative to large banks in state (%)	1.39	-0.19	-0.07
Indicator if governor is Democrat	0.55	0.05	-0.01
Share of state legislature that is Democrat	0.65	0.05	0.04
State annual personal income growth: 1993 to 1997	0.053	0.32**	0.33**
State annual personal income growth: 1998 to 2002	0.051	-	0.07

Table IV: Correlations between state characteristics and the branching restrictions index.

Read Table V:

- Table V reestimates the interest-rate models after adding possible determinants of the branching restrictions index.
- The branching index remains positive and statistically meaningful in 1998 and 2003 across panels.
- The robustness check shows that the price result is not absorbed by small-bank share, insurance-sector size, small-firm share, relative bank capital, governor party, or legislature party.

Economic message: Observable state political-economy variables help explain regulation, but they do not explain away the loan-rate effect.

Table V
Regression of Interest Rate on Most Recent Loan on State, Bank, and Borrower Characteristics
Controlling for Possible Determinants of the Branching Restrictions Index

Each panel reports regression results that add one of the possible determinants of the branching index (from Table IV) to the interest rate models from Table III. These regressions include the same set of other variables. A complete set of results is available in the Internet Appendix. The pooled model is not estimated because the state characteristics are not identified with state fixed effects. In the 1993 sample, we assign the 2003 value for the branching index. Standard errors are clustered by state.

	1993 (Control Group)		1998		2003	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
Panel A						
Index of branching restrictions (4 is most, 0 least, restricted)	0.016	0.23	0.250	3.64	0.266	3.50
Small bank share of all banking assets in state	-0.011	0.43	-0.049	1.46	-0.036	1.00
N	1,636		788		1,737	
R ²	16.32%		20.32%		23.60%	
Panel B						
Index of branching restrictions (4 is most, 0 least, restricted)	0.010	0.15	0.210	2.93	0.244	3.75
Relative size of insurance to banking plus insurance	0.004	0.40	-0.009	0.60	-0.025	1.77
N	1,636		788		1,737	
R ²	16.33%		20.14%		23.87%	
Panel C						
Index of branching restrictions (4 is most, 0 least, restricted)	0.016	0.25	0.175	2.41	0.230	3.49
Small firm share of the number of firms in the state	-0.058	1.59	0.139	2.04	0.138	2.47
N	1,636		788		1,737	
R ²	16.46%		20.85%		24.10%	

(continued)

Table V, part 1: Adding possible determinants of state restrictions.

Table V—Continued

	1993 (Control Group)		1998		2003	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
Panel D						
Index of branching restrictions (4 is most, 0 least, restricted)	0.010	0.15	0.217	2.87	0.254	3.65
Capital ratio of small banks relative to large banks in state (%)	0.002	0.36	-0.008	0.38	-0.011	1.28
N	1,636		788		1,737	
R ²	16.31%		20.06%		23.56%	
Panel E						
Index of branching restrictions (4 is most, 0 least, restricted)	0.010	0.16	0.207	2.80	0.266	3.74
Indicator if governor is Democrat	0.015	0.09	0.137	0.55	0.315	1.22
N	1,636		788		1,737	
R ²	16.31%		20.13%		23.72%	
Panel F						
Index of branching restrictions (4 is most, 0 least, restricted)	0.003	0.04	0.262	3.89	0.263	3.88
Share of state legislature that is Democrat	0.076	0.42	-0.474	1.55	-0.115	0.56
N	1,636		788		1,737	
R ²	16.33%		20.67%		23.53%	

Table V, part 2: Continued robustness checks.

6.5 Table VI: Bank debt, total debt, denial, and trade credit

Read:

- Table VI studies whether lower borrowing costs translate into more borrowing or fewer credit constraints.

- The coefficient on the branching index is negative for the bank-debt indicator. Since a higher index means more restrictions, this implies firms are more likely to borrow from banks in more open states.
- The branching index is not statistically important for log total debt.
- The index is not important for the indicator that a firm was denied or discouraged from applying for a loan.
- The late trade-credit result is suggestive but not central: the evidence is consistent with the possibility that competition can coexist with, or even slightly increase, credit rationing.

Economic message: Openness shifts firms toward bank borrowing, but it does not produce a broad increase in total debt or a clear reduction in credit denial.

Table VI
Regression of Log Total Debt; Probability of Denial; and Late Trade Credit on State, Bank, and Borrower Characteristics (Pooled Models)

This table reports regressions of financial policies by small firms, weighted by the survey weights to account for disproportionate sampling and unit nonresponse, and includes a set of two-digit SIC indicator variables to control for industry effects. Standard errors are clustered by state-year. A Probit model is estimated for the bank debt and denied/discouraged regressions. The marginal effects, rather than the Probit coefficients, are reported. Market concentration equals the sum of squared share of deposits held by all banks in the borrower's local market, where market is defined as the Metropolitan Statistical Area (MSA) or county. Standard errors are clustered by state-year.

	Indicator if Firm Has Bank Debt		Log of Total Debt		Indicator if Firm Was Denied or Discouraged from Applying for a Loan		Percent Late Payments on Trade Credit	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
Index of branching restrictions (4 is most, 0 least, restricted)	-0.015	1.96	0.045	0.86	-0.002	0.37	-0.542	1.79
<i>Borrower characteristics</i>								
Log of borrower assets	0.043	9.73	0.556	14.89	-0.023	7.39	-0.063	0.29
Indicator if borrower is a corporation	-0.041	2.35	0.132	1.04	0.014	1.23	0.890	1.08
Borrower ROA	0.008	1.34	0.139	2.74	-0.024	4.89	-0.355	0.91
Borrower risk rating (1 is safest; 5 is riskiest)	0.013	2.24	-0.059	1.32	0.077	16.54	3.425	8.64
<i>Relationship variables and age</i>								
Length of longest bank relationship	-0.001	1.37	-0.002	0.31	-0.003	3.78	0.046	1.05
Borrower age (years)	0.000	0.09	-0.015	3.02	-0.003	5.21	-0.043	1.50
Number of relationships	0.320	21.79	1.181	23.79	0.053	11.89	1.334	4.99
<i>Market characteristics</i>								
Market concentration	0.081	1.05	0.347	0.55	-0.076	1.16	-0.122	0.03
Indicator if borrower is in an MSA	-0.079	4.38	-0.326	2.16	0.029	1.81	1.129	1.46
Average lagged 5-year growth rate	0.172	1.60	2.020	2.27	-0.043	0.59	-1.075	0.24
N	11,945		11,957		11,939		8,236	
R ²	34.04%		22.59%		12.48%		12.65%	

Table VI: Total debt, bank debt, denied credit, and late trade-credit outcomes.

6.6 Table VII: Nonprice loan terms

Read:

- Table VII studies collateral, guarantees, and maturity for the most recent loan.
- The branching index does not significantly predict whether the loan has collateral.
- The branching index does not significantly predict whether the loan is guaranteed.
- The branching index does not significantly predict loan maturity.
- Several borrower and relationship controls matter, but the state branching regulation variable does not appear to loosen nonprice loan terms.

Economic message: Competition lowers loan rates, but the structure of the loan contract remains largely unchanged. This is consistent with persistent information and enforcement frictions in small-firm lending.

TABLE VII
Regression of Nonprice Loan Terms from Most Recent Loan on State, Bank, and Borrower Characteristics (Pooled Models)

This table reports regressions weighted by the survey weights to account for disproportionate sampling and unit nonresponse, and includes a set of two-digit SIC indicator variables to control for industry effects. Standard errors are clustered by state-year. A Probit model is estimated for the collateral and guarantee regressions. The marginal effects, rather than the Probit coefficients, are reported. Market concentration equals the sum of squared share of deposits held by all banks in the borrower's local market, where market is defined as the Metropolitan Statistical Area (MSA) or county. Standard errors are clustered by state-year.

	Indicator if Loan Has Collateral		Indicator if Loan Is Guaranteed		Loan Maturity (Years)	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
Index of branching restrictions (4 is most, 0 least, restricted)	-0.011	0.87	0.002	0.15	0.198	1.38
<i>Interest rates</i>						
Prime interest rate	-0.029	0.31	0.095	1.12	-0.123	0.17
Term structure (10-year government bond - 3-month T-bill)	-0.005	0.13	0.000	0.00	-0.464	1.36
Default premium (BAA - 10-year government bond)	-0.066	1.76	-0.035	1.11	-0.442	1.47
<i>Borrower characteristics</i>						
Log of borrower assets	0.053	6.30	0.022	2.73	0.188	1.72
Indicator if borrower is a corporation	-0.038	1.53	0.089	3.40	-1.172	4.96
Borrower ROA	-0.010	0.57	-0.025	1.93	-0.280	1.72
Borrower risk rating (1 is safest; 5 is riskiest)	0.024	2.62	0.024	2.37	0.104	0.81
<i>Loan characteristics</i>						
Indicator if loan is floating-rate	-0.023	1.02	0.117	4.53	0.057	0.22
<i>Lender characteristics</i>						
Indicator if lender is a bank	0.052	1.21	0.092	2.03	0.146	0.25
Indicator if lender is a nonfinancial company	-0.131	1.32	0.058	0.55	-0.773	0.75
<i>Relationship variables</i>						
Length of lender-borrower relationship (years)	-0.001	0.40	-0.004	3.19	-0.023	1.58
Borrower age (years)	-0.001	0.89	-0.002	1.70	0.005	0.37
Number of information services from lender	0.017	0.52	-0.015	0.45	-0.775	3.39
Number of noninformation services from lender	0.014	0.40	0.023	0.98	-0.755	3.17
Indicator if borrower has a deposit with lender	-0.110	3.67	0.032	1.00	-1.586	4.12
Number of relationships	0.009	1.17	0.039	4.41	-0.285	2.90

Table VII, part 1: Nonprice loan terms.

Table VII—Continued

	Indicator if Loan Has Collateral		Indicator if Loan Is Guaranteed		Loan Maturity (Years)	
	Coefficient	t-stat	Coefficient	t-stat	Coefficient	t-stat
<i>Market characteristics</i>						
Market concentration	0.077	0.64	-0.062	0.49	-0.935	0.75
Indicator if borrower is in an MSA	0.008	0.27	0.019	0.70	-0.359	1.20
Average lagged 5-year growth rate	-0.085	0.45	0.232	1.36	-0.465	0.28
<i>N</i>	4,153		4,156		4,065	
<i>R</i> ²	11.86%		10.02%		15.91%	

Table VII, part 2: Continued nonprice loan terms.

7 Short summary

- IBBEA allowed interstate banking and branching, but states retained authority to restrict several forms of out-of-state entry.
- Rice and Strahan build a state-year branching restrictions index from zero to four, where higher values mean more barriers.
- They merge this index with the 1993, 1998, and 2003 Surveys of Small Business Finance.
- The 1993 survey is a pre-treatment falsification sample; 1998 and 2003 are post-treatment samples.
- Firms in more restrictive states pay higher loan interest rates after IBBEA.
- The implied rate difference between most-open and most-restricted states is about 80 to 100 basis points.
- Firms in open states are more likely to borrow from banks.
- There is little evidence that open states experience higher total debt, lower loan denial, longer maturity, less collateral, or fewer guarantees.
- The paper therefore finds a strong price response but a weak quantity response.

- The results suggest that competition expands credit supply through prices but does not necessarily eliminate small-firm credit frictions.

8 Conclusion

8.1 Core conclusion

State openness to interstate branching lowered the cost of small-business borrowing. After interstate branching became effective, firms in states with fewer branching barriers paid substantially lower loan rates than similar firms in more restrictive states.

8.2 Economic intuition

Entry by out-of-state banks increases competitive pressure on incumbent lenders. Banks respond by reducing loan rates and serving more small firms through bank debt. But information and contracting frictions limit the ability of competition to expand quantities.

8.3 Incremental contribution revisited

The paper contributes by showing that a competition-driven credit-supply expansion operates differently from a bank-liquidity shock. It clearly lowers loan prices, but does not obviously increase total borrowing or loosen contract terms.

8.4 Implications

- Banking competition can benefit small firms through lower interest rates.
- Lower loan prices do not necessarily imply that credit constraints have disappeared.
- Policies that increase competition may need to be complemented by institutions that improve screening, information sharing, collateral enforcement, and relationship formation.
- The effect of credit supply on capital structure depends on the type of supply shock.

8.5 Limitations

- The SSBF is a repeated cross section, not a firm panel.
- State branching rules are policy choices, not randomly assigned treatments.
- Loan-rate data are only available for firms with a most recent loan, so the price regressions condition on borrowing.
- The branching index is a regulatory proxy for competition, not a direct measure of all local competitive forces.

8.6 Takeaway

The paper's main lesson is that credit competition matters, but it matters primarily through loan prices. Open banking markets make credit cheaper and encourage bank borrowing, yet they do not necessarily make small firms borrow more or receive looser contracts.